

The Product Design Process: From Observations to Opportunities

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Agenda

1. Greenhouse Debrief & Synthesis
2. The Product Design Process Framework
3. From Observations to Opportunities
4. Final Project Introduction

Part 1: Greenhouse Debrief & Synthesis

Welcome Back!

You spent some time at the greenhouse facility

You know a bit more about greenhouse operations

Today's goal:

Turn your observations into **actionable design opportunities**.

Activity 1: Gallery Walk (20 min)

Setup:

- In a table, each group shows 2–3 key observations (on paper or screen)
- Include: photos, sketches, problem statements, quotes

Your task:

- Walk around and read other groups' observations
- Use sticky notes to add:
 -  "We saw this too"
 -  "Tell us more"
 -  "This could connect to..."

Activity 2: Pattern Recognition

Class discussion:

Let's identify **patterns** across all observations:

- What problems appeared in multiple groups' notes?
- What themes emerged?
- What surprised everyone?
- What assumptions were challenged?

The Big Question

You have observations.

You have problems.

You have ideas.

Now what?

How do professionals turn observations into products?

Part 2: The Product Design Process Framework

Why Do We Need a Process?

Without process:

- ✗ Jump to solutions too quickly
- ✗ Miss important user needs
- ✗ Waste resources on wrong features
- ✗ Fail to coordinate teams
- ✗ Can't measure progress

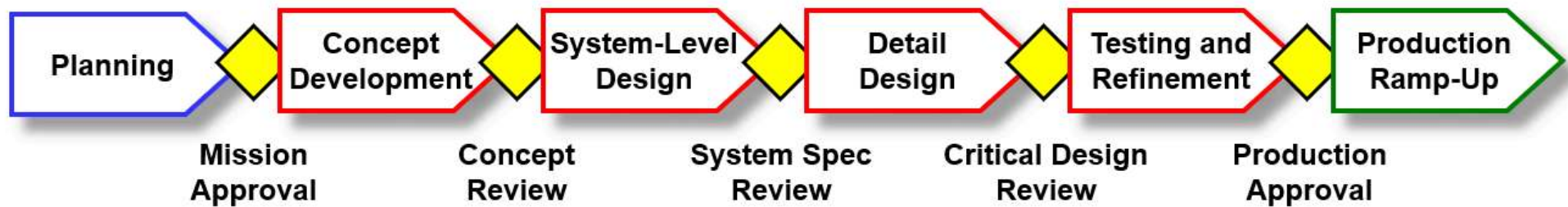
Why Do We Need a Process? (cont.)

With process:

-  Systematic exploration of alternatives
-  Reduced risk
-  Better team coordination
-  Clear milestones and decision points

The Generic Product Development Process

Based on Ulrich, Eppinger & Yang, Chapter 2



Phase 0: Planning

Activities:

- Identify market opportunities
- Define business strategy
- Assess technical feasibility
- Allocate resources

Output: Project mission statement

Greenhouse example: "We noticed workers struggle with X. Is this worth solving?"

Phase 1: Concept Development

Activities:

- Identify customer needs
- Establish target specifications
- Generate product concepts
- Select product concept(s)
- Test concept(s) with customers

Output: Concept definition, specifications

Greenhouse example: "Workers need a hands-free way to record observations."

Phase 2: System-Level Design

Activities:

- Define product architecture
- Decompose into subsystems and components
- Preliminary design of key components
- Identify suppliers for key components

Output: System architecture, component specifications

Greenhouse example: "Our system has 3 subsystems: sensing, data storage, user interface."

Phase 3: Detail Design

Activities:

- Complete specification of geometry, materials, tolerances
- Identify all standard and custom parts
- Establish process plans
- Create tooling designs

Output: Complete CAD models, drawings, specifications

Greenhouse example: "The housing is ABS plastic, 3mm thick, with IP65 rating."

Phase 4: Testing and Refinement

Activities:

- Build and evaluate prototypes
- Test performance against specifications
- Identify and resolve problems
- Iterate design

Output: Validated design, test results

Greenhouse example: "We tested in 90% humidity for 48 hours. The seal failed. Redesign is needed."

Phase 5: Production Ramp-Up

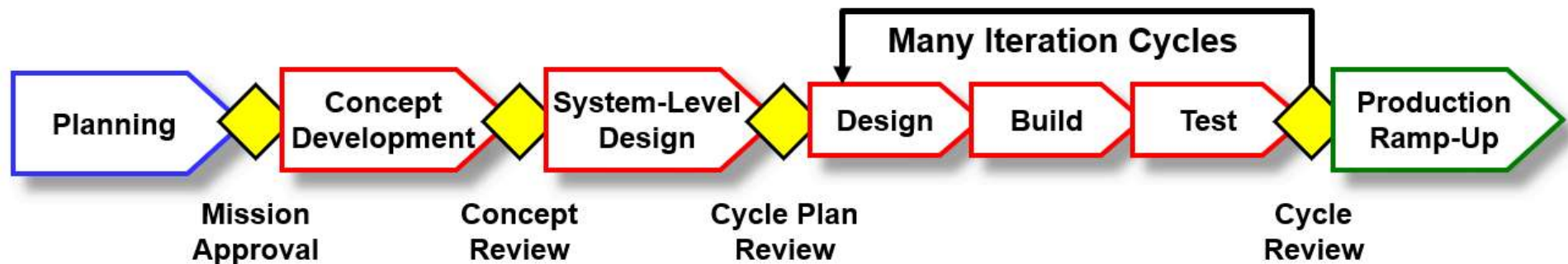
Activities:

- Train workforce
- Refine production process
- Evaluate early production units
- Transition to full production

Output: Product ready for market

Greenhouse example: "We can manufacture 100 units per week with our assembly process."

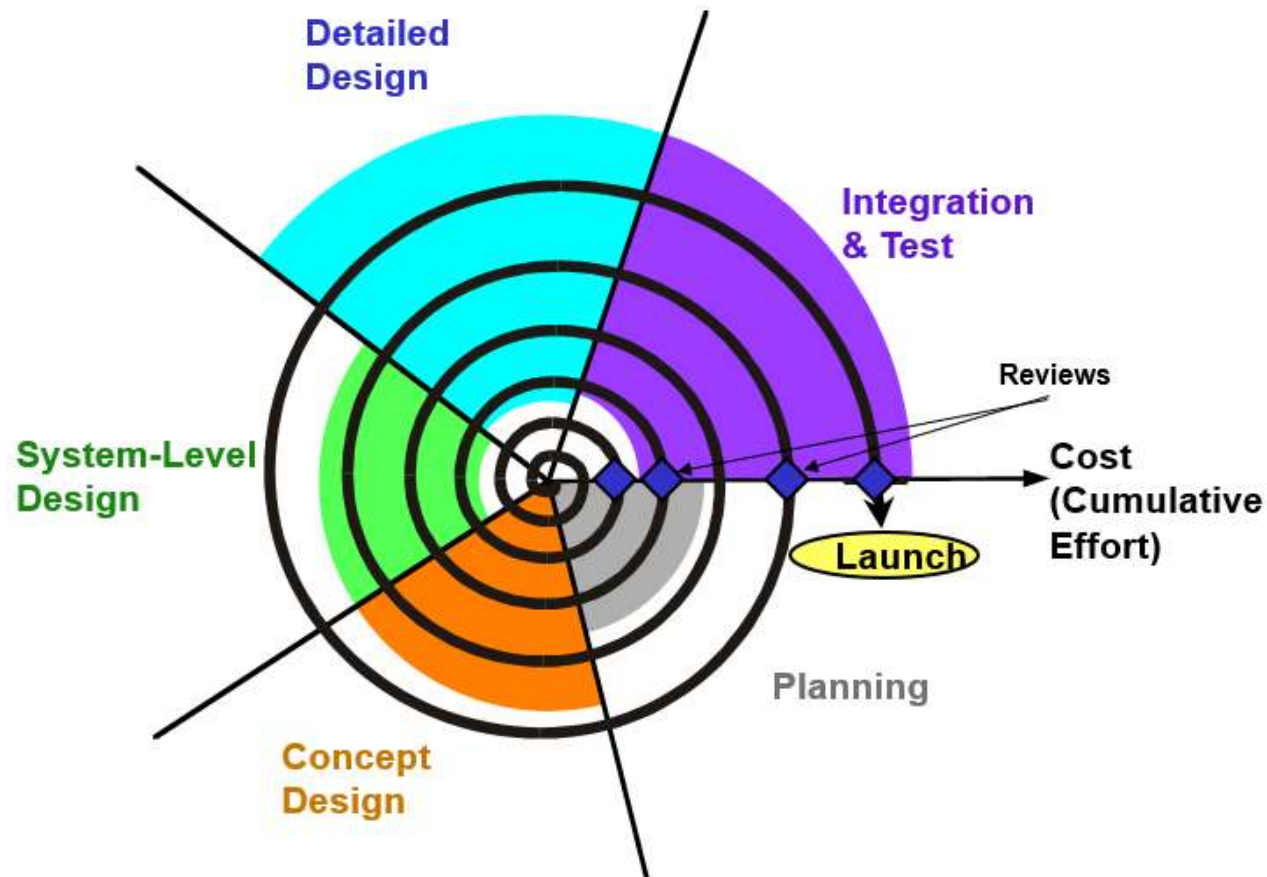
The Process is Iterative (1)



Key insight:

- You don't go through phases exactly once
- Learning in later phases informs earlier phases
- Iteration is expected and necessary

The Process is Iterative (2)

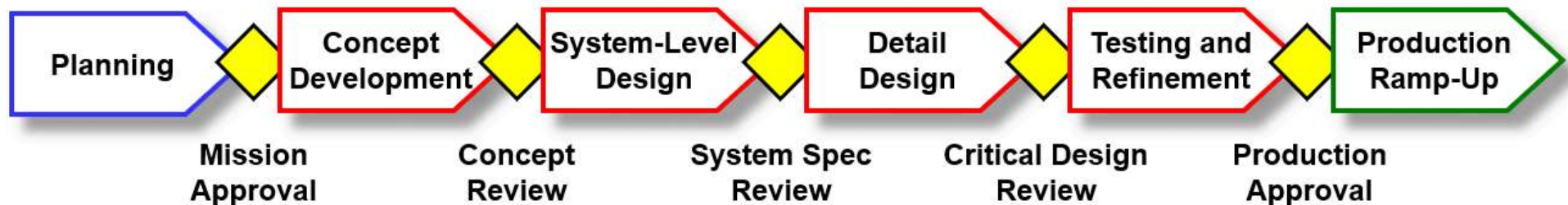


Stage-Gate Process

Alternative view: Product development as a series of **stages** and **gates**.

Stage: Work phase (concept development, testing, etc.)

Gate: Decision point (Go / Kill / Hold / Recycle)



Why Gates Matter

Gates are decision points where you ask:

- Is this opportunity still worth pursuing?
- Do we have the resources to continue?
- Does the design meet our criteria?
- Should we pivot or proceed?

Example gates:

- Gate 1: Opportunity assessment / Gate 2: Concept feasibility
- Gate 3: Design validation / Gate 4: Production readiness

Process Variations by Industry

Industry	Characteristics
Consumer electronics	Fast cycles, high iteration, cost-driven
Medical devices	Heavy regulation, long validation, safety-critical
Robotics	Hardware-software integration, field testing
Startups	Lean, MVP-focused, rapid pivots

Key point: The process adapts, but the **phases remain**.

Minimum viable product (MVP) is a version of a product with just enough features to be usable by early customers who can then provide feedback for future product development ([Wikipedia](#)).

Your Initial Challenge: Process Mapping

Let's map your visibility light challenge to the process.

What did you actually do?

Phase	What You Did	Maybe What You Skipped
Planning	Read the brief	Market research, user research
Concept Development	Brainstormed ideas	Systematic concept generation, testing
System-Level Design	Sketched architecture	Formal decomposition
Detail Design	Built prototype	Specifications, tolerances
Testing	Turned it on	User testing, durability testing
Production	Made 2 units	Process planning, scaling

Key Insight

You compressed 6 phases into 2 weeks. That's why it was hard.

Real products take months or years to develop. Check the figure in the next page, from Ulrich and Eppinger.

The **process** helps manage complexity and reduce risk.

	Belle-V Ice Cream Scoop	AvaTech Avalanche Probe	iRobot Roomba Vacuum Cleaner	Tesla Model S Automobile	Boeing 787 Aircraft
Annual production volume	10,000 units/year	1,000 units/year	2,000,000 units/year	50,000 units/year	120 units/year
Sales lifetime	10 years	3 years	2 years	5 years	40 years
Sales price	\$40	\$2,250	\$500	\$80,000	\$250 million
Number of unique parts (part numbers)	2 parts	175 parts	1,000 parts	10,000 parts	130,000 parts
Development time	1 year	2 years	2 years	4 years	7 years
Internal development team (peak size)	4 people	6 people	100 people	1000 people	7,000 people
External development team (peak size)	2 people	12 people	100 people	1000 people	10,000 people
Development cost	\$100,000	\$1 million	\$50 million	\$500 million	\$15 billion
Production investment	\$20,000	\$250,000	\$10 million	\$500 million	\$15 billion

EXHIBIT 1-3 Attributes of five products and their associated development efforts. All figures are approximate, based on publicly available information, estimates, and company sources.

Part 3: From Observations to Opportunities

The Front End of Innovation

The "fuzzy front end" is the earliest phase of product development.

Characteristics:

- Highest uncertainty
- Most critical for success
- Defines what product you'll build

Key activities:

1. Opportunity identification  (hopefully you did this at the greenhouse)
2. Opportunity assessment  (we'll do this now)
3. Concept generation (next weeks)

Not All Opportunities Are Equal

You identified many problems at the greenhouse.

But:

- Some are more important than others
- Some are more feasible than others
- Some are better fits for your skills and timeline

We need a systematic way to evaluate opportunities.

Activity 3: Opportunity Assessment Matrix (25 min)

Setup:

Take your **3 problem statements** from the greenhouse report.

Task:

Rate each opportunity on a scale of 1–5 using the matrix on the next slide.

Outcome:

Identify your **top opportunity** to pursue for the final project.

Opportunity Assessment Matrix

Criterion	Problem 1	Problem 2	Problem 3
Impact (How much does it matter to users?)			
Frequency (How often does this problem occur?)			
Feasibility (Can we solve it with our skills?)			
Scope (Can we prototype it in 4 weeks?)			
Interest (Does our team care about this?)			
TOTAL			

Scale: 1 = Low, 5 = High

Scoring Guidance

Impact: Would solving this make a significant difference?

Frequency: Daily problem = 5, Rare problem = 1

Feasibility: Do you have the technical skills and resources?

Scope: Can you build a minimum prototype in 4 weeks?

Interest: Are you excited to work on this?

Calculate total score for each problem.

The highest score is your top opportunity (but use judgment too).

Class Discussion

Let's hear from the groups:

- What's your top opportunity?
- What was your rationale?
- What concerns do you have?

Check scope and feasibility

Different Opportunities, Different Processes

Example 1: Environmental monitoring system

→ Heavy on sensing and IoT, lighter on mechanical design

Example 2: Automated harvesting tool

→ Heavy on robotics and manipulation, complex testing

Example 3: Ergonomic tool redesign

→ Heavy on user testing and ergonomics, simpler electronics

The process adapts to the product type.

Part 4: Final Project Introduction

Final Project: Greenhouse Product Design Challenge

Objective:

Design a product that addresses a real need identified during the greenhouse visit.

Timeline: Weeks 5–8 (4 weeks)

Groups: max. 4 students, min. 2 nationalities

Constraints

- Must address a **real problem** from your observations
- Must be **feasible to create a mockup** in 3 weeks
- Must demonstrate **one working subsystem**

Deliverables

Week 5: User requirements document

Week 6: Concept selection and system architecture

Week 7: Mockup/prototype and tests

Week 8: Final presentation and demonstration

Assessment Criteria

Criterion	Weight
Problem definition and user research	20%
Design process and documentation	50%
Prototype quality and innovation	20%
Presentation and demonstration	10%

We value the **process**

Next Steps

This week:

1. Finalize your opportunity selection
2. Do a market research (competitors, patents)
3. Begin deeper user research

Due before Week 5 lecture:

- Project proposal (template in Learn/Moodle)

Key Takeaways from Week 4

1. **Product development is a structured process** with distinct phases
2. **The fuzzy front end** (opportunity identification) is critical
3. **Not all opportunities are equal** — assess systematically
4. **Your final project** will follow this process over the next weeks

Thank you!

Next week: Identifying User Needs